

Novel Substrate with Combined Embedded Capacitance and Resistance for Better Electrical Performance and Higher Integration

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Agenda

- **Introduction**
- **Embedded Passive Drivers**
- **R/C Material Properties**
- **Processing**
- **Measurements**
- **Applications**
- **Future Studies**
- **Conclusions**

Introduction

- **Growing number of PCBs with embedded passives**
- **Drivers: High Frequencies and Miniaturization**
- **Embedding Resistors within existing layers**
- **Embedding Capacitance between layers**
- **Many PCBs with both Embedded Resistors and Capacitors in separate cores**
- **Technical advantages combining Embedded Resistors and Capacitors in same core**
- **Economic advantages of a combined Resistor/Capacitor (R/C) material**

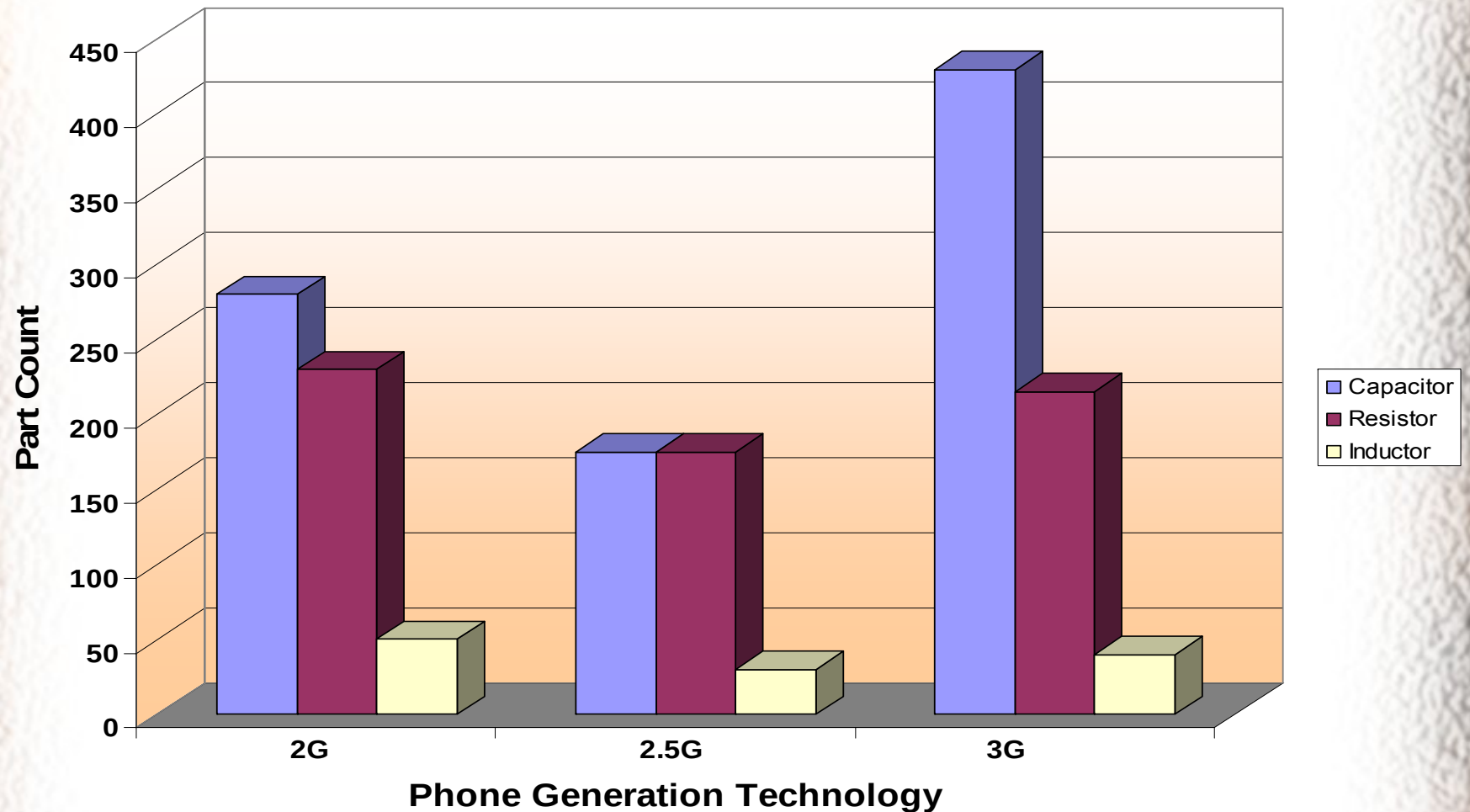
Embedded Passives Drivers: Increasing Number of Discrete Components

System	Total Passives	Total Ics	Ratio
CELLULAR PHONES			
Ericsson DH338 Digital	359	25	14:1
Ericsson E237 Analog	243	14	17:1
Philips PR93 Analog	283	11	25:1
Nokia 2110 Digital	432	21	20:1
Motorola Md 1.8 GHz	389	27	14:1
Casio PH-250	373	29	13:1
Motorola StarTAC	993	45	22:1
Matsushita NTT DOCOMO 1	492	30	16:1
CONSUMER PORTABLE			
Motorola Tango Pager	437	15	29:1
Casio QV10 Digital Camera	489	17	29:1
1990 Sony Camcorder	1226	14	33:1
Sony Handy Cam DCR-PC7	1329	43	31:1
OTHER COMMUNICATION			
Motorola Pen Pager	142	3	47:1
Infotac Radio Modem	585	24	24:1
Data Race Fax-Modem	101	8	13:1
PAD			
Sony Magic Link	538	74	7:1
COMPUTERS			
Apple Portable Logic Board	184	24	8:1
Apple G4	457	42	11:1

Over 20:1 Ratio of Passives/IC is typical !

Embedded Passives Drivers: Increasing Number of Discrete Components

EMBEDDED PASSIVE PARTS IN PHONE



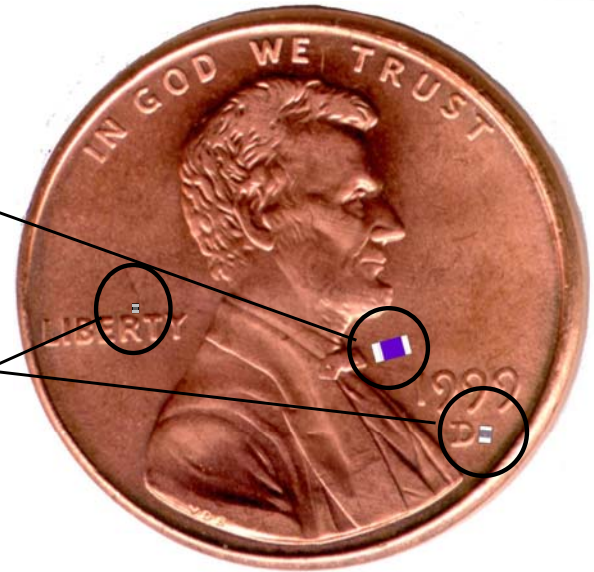
Embedded Passives Drivers: Shrinking Discrete Components

0402 resistor:

- (It's the same size as President Lincoln's bow tie)

Now find the 0201 capacitor:

**Is there such a thing as
an 01005?:**



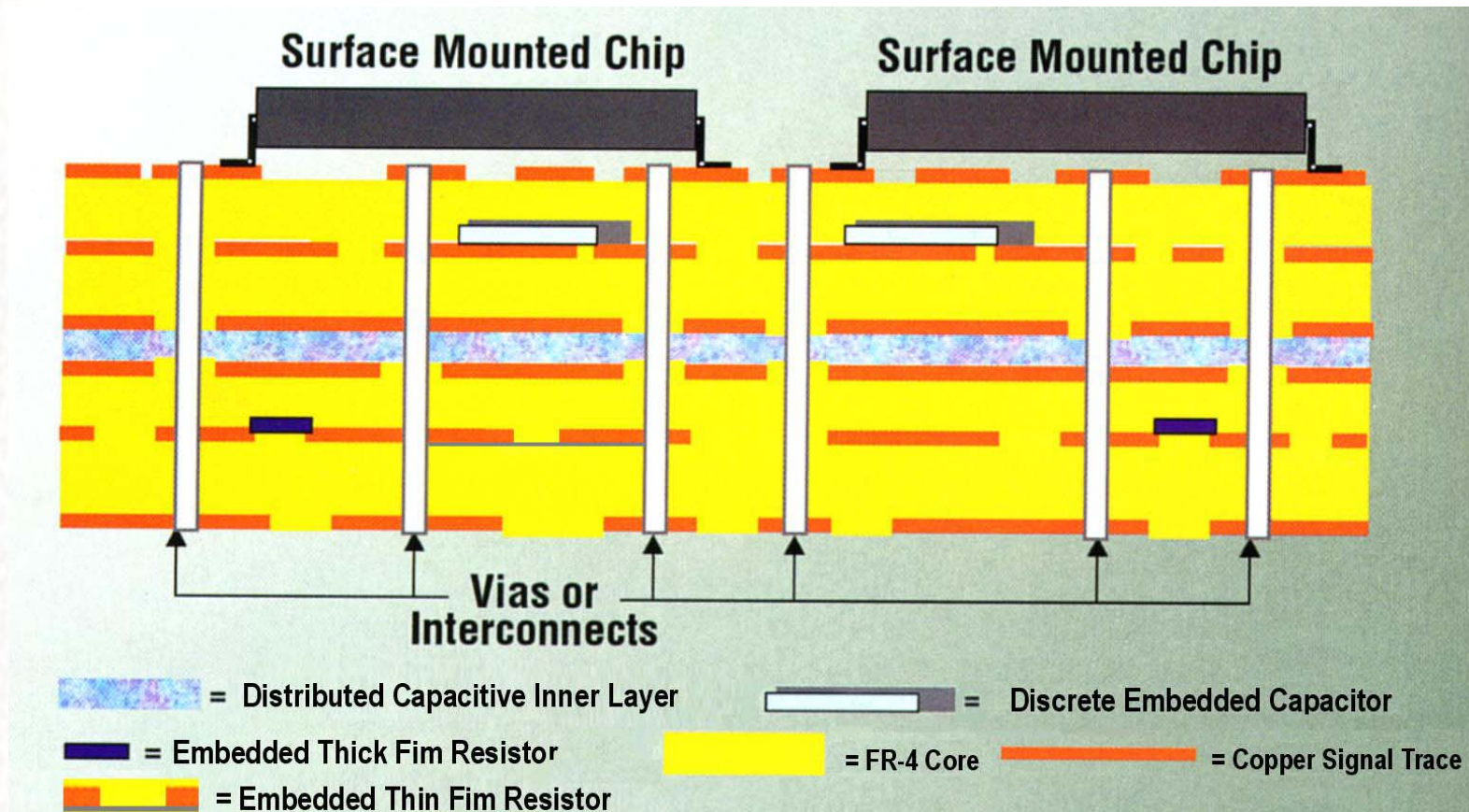
- **Now inspect 2000 parts like this on one board**
- **Still need to connect through 2 Vias**

Courtesy of



*Bristlecone
Wisdom of the Ages*

Cross-Section View of PCB Board Containing Embedded Passive Components



Courtesy of Dr. B. Borland

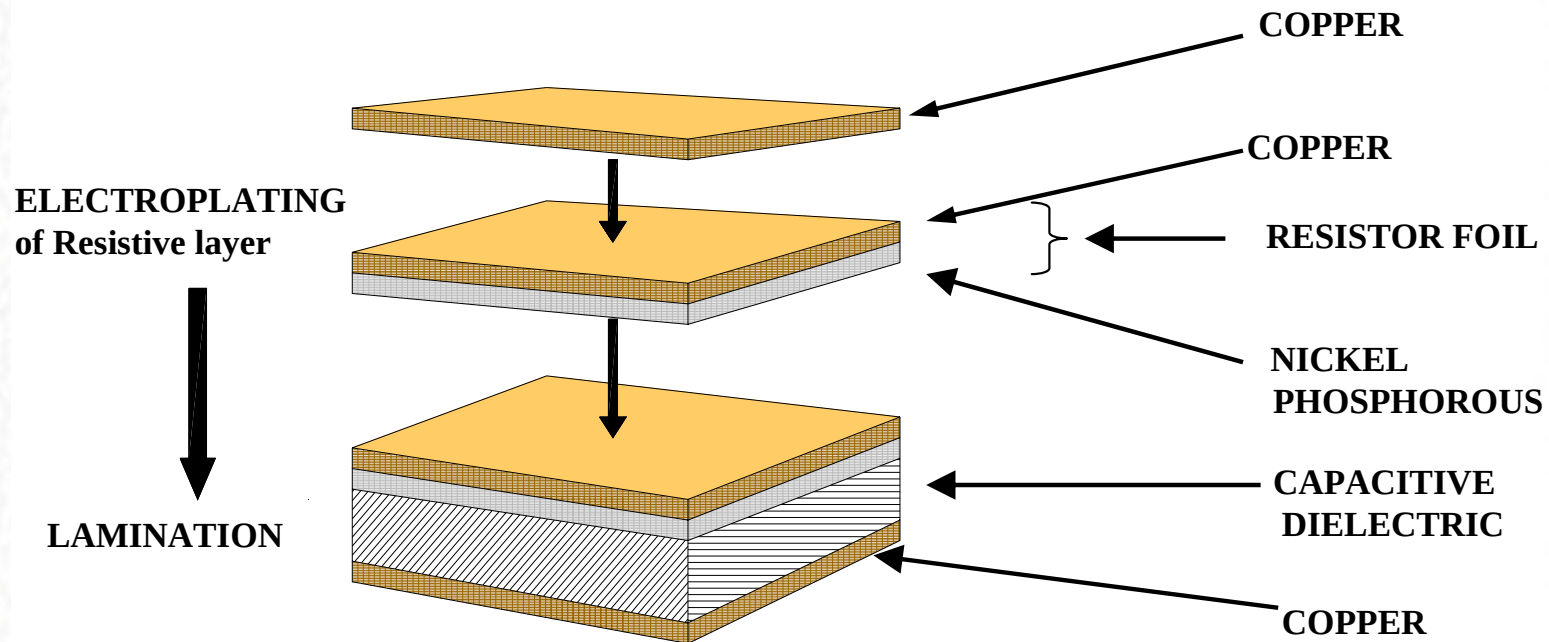
Embedded Resistor/Capacitor Technology

Benefits

- **Combined Laminate Product-Resistance and Capacitance in the same core.**
- **Developed to accommodate high density designs.**
- **Embedded Resistor and R/C Networks**
- **Improve signal integrity by better impedance matching.**
- **Improve signal to noise ratios.**
- **Improve Power Distribution**
- **Reduce EMI**
- **Standard PCB Subtractive Processing.**
- **Better cost effectiveness than separate Resistor and Capacitor cores.**

Resistive/Capacitive Material Properties- Overview

- R/C material is a combined product of a thin film NiP alloy electrodeposited on copper foil that is laminated to a thin dielectric material
- Subtractively processed to produce embedded R/C Networks.



Resistive/Capacitive Material Properties-

Resistor Foil Sheet Resistivity vs. Film Thickness

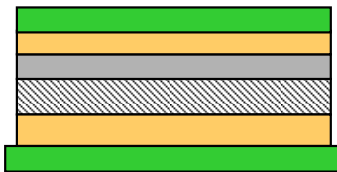
Sheet Resistivity	NiP Film Average Thickness	Material Tolerance
10 $\Omega/\square$	1.00 Micron	3%
25 $\Omega/\square$	0.40 Micron	5%
50 $\Omega/\square$	0.20 Micron	5%
100 $\Omega/\square$	0.10 Micron	5%
250 $\Omega/\square$	0.05 Micron	10%

Resistor/Capacitor Material Properties

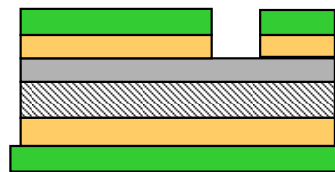
Properties	Method	Ohmega/FaradFlex Core
Copper Weight, μm	Nominal	35
Sheet Resistivity, ohms / square	Nominal	10-250
Dielectric Thickness, μm	Nominal	24
Cp@ 1MHz, nF/in ² (pF/cm ²)	IPC-TM 650 2.5.5.3	1.0 (155)
Dk @1MHz	IPC-TM 650 2.5.5.3	4.4
Loss Tangent @ 1MHz	IPC-TM 650 2.5.5.3	0.015
Peel Strength, lbs/in	IPC-TM 650 2.4.9	5.0
Dielectric Strength, kV/mil	IPC-TM 650 2.5.6.3	5.3
Tensile Strength, Mpa(kpsi)	ASTM D-882 A	152(22.0)
Elongation, %	ASTM D-882 A	18.5

Processing of R/C Material

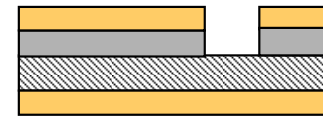
STEP 1:
Apply Photoresist to
Laminate on both
sides



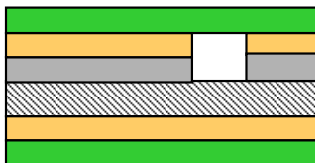
STEP 2:
Image and Etch Copper on
R-Foil Using Any
Conventional
Etchant (1st etch)



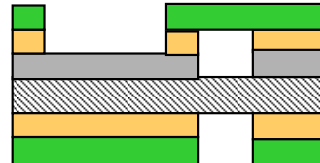
STEP 3:
Etch Unwanted Resistive
Material with Copper Sulfate
Solution
(2nd etching process)



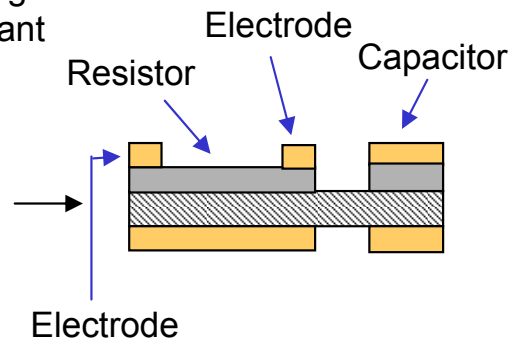
STEP 4:
Apply Photoresist to
Laminate on both
sides (Vacuum)



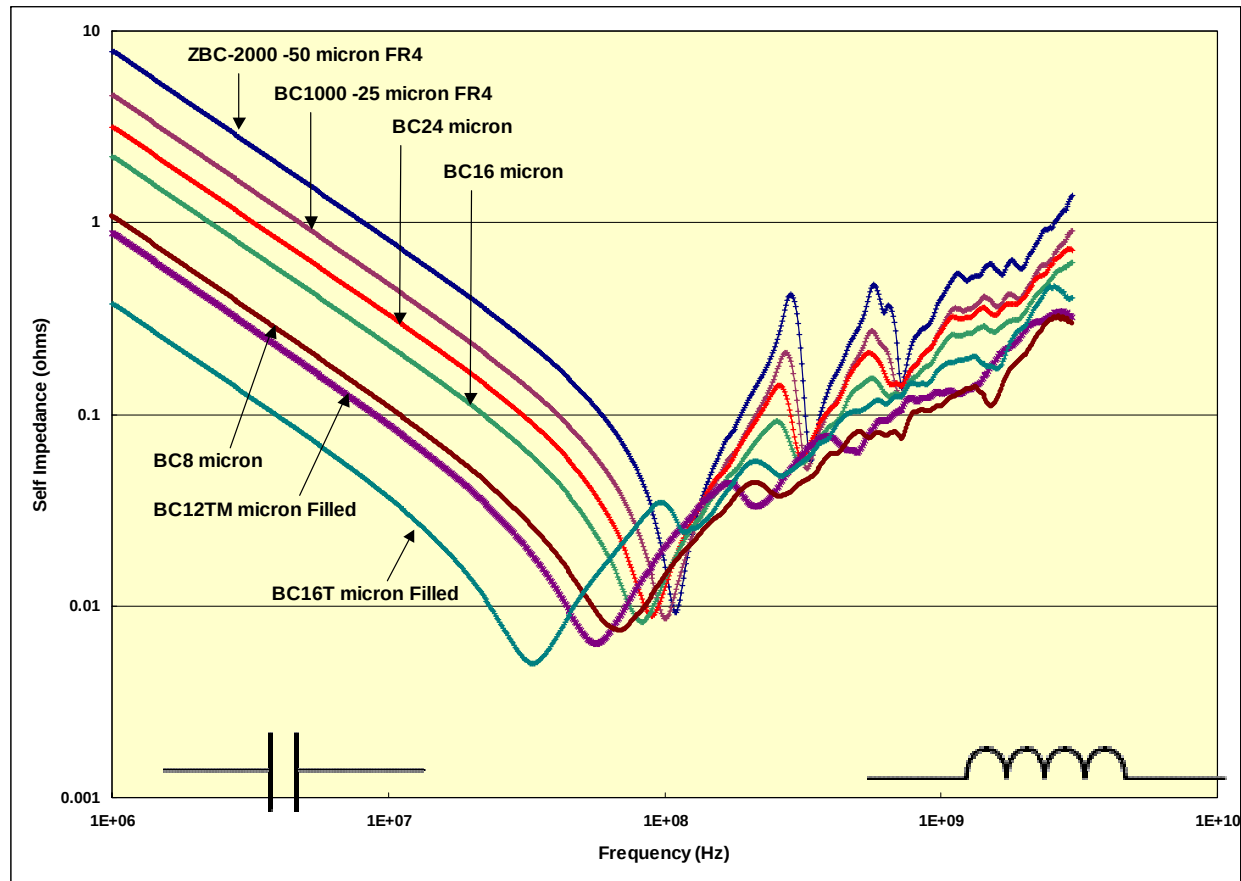
STEP 5:
Etch Away Copper over the
Designed Resistor Using a
Selective Alkaline Etchant
(3rd etch)



STEP 6:
Strip Photoresist



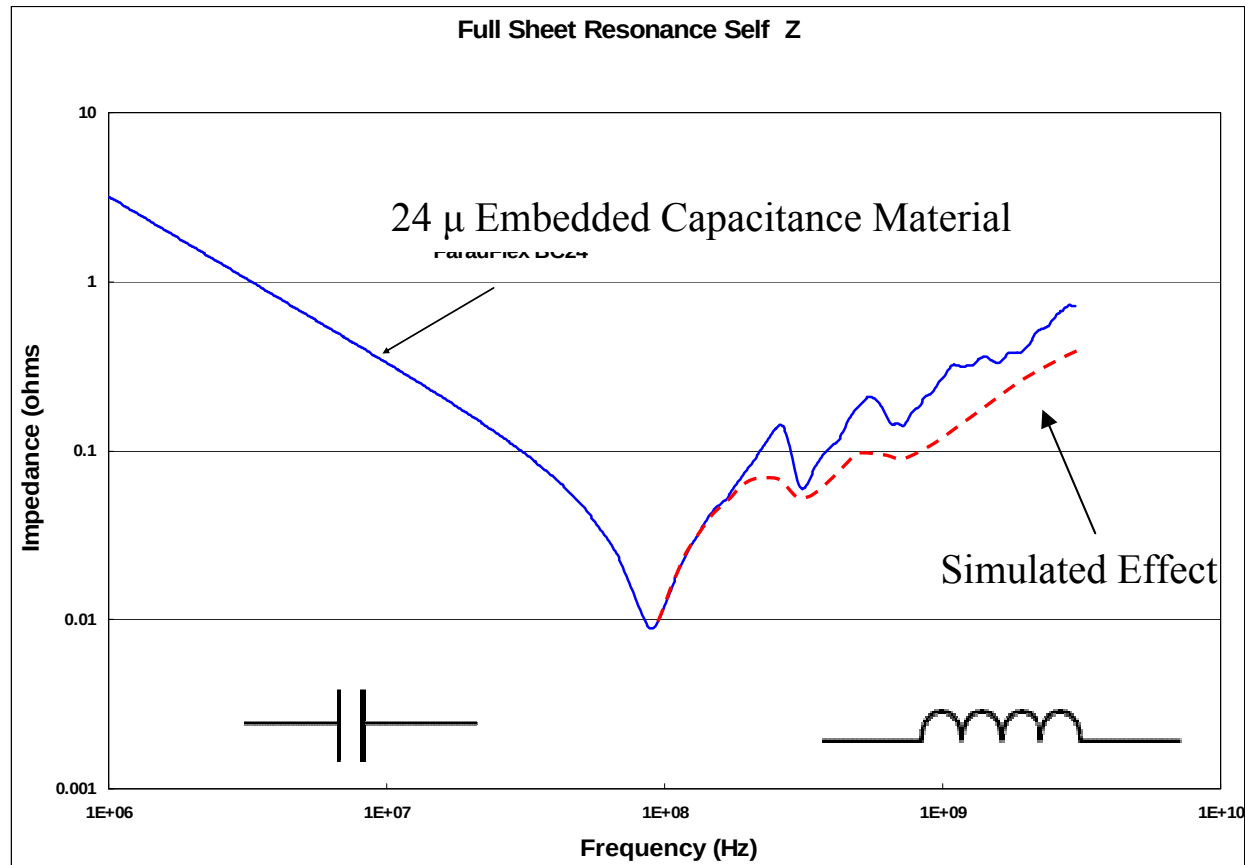
Measurements- Electrical Performance Capacitor with Cu/Cu



Thinner is better for Reducing Resonance

Measurements- Electrical Performance Capacitor

Effect of Increased Loss on Impedance of Capacitance Material



Increased Loss due to Resistor layer makes Capacitance Material perform as if it is Thinner!

Measurements- Electrical Performance

Resistor Layer

Capacitor Material vs. FR4

Properties	NiP/Capacitor Core	NiP Core FR-4 (control)	Remarks and Conditions
Sheet Resistivities (ohm/square)	25	25	Nominal
Material Tolerance	+/-5%	+/-5 %	
Load Life Cycling Test Resistor Size: 0.500" X 0.050" Loaded: (Δ R%) @ 150mW Unloaded: (Δ R%)	<0.9 after 3200 hrs.) <0.74 after 3200 hrs.)	<5	MIL-STD-202-108I Ambient Temp: 70C On Cycle: 1.5 hrs Off Cycle: 1.5 hrs Length Of Test: 10000 hrs
Current Noise Index in dB	<-23	<-15	MIL-STD-202-308 Voltage Applied: 5.6 Volts
Humidity Test (Δ R%)	0.5	0.5	MIL-STD-202-103A Temp: 40 °C Relative Humidity: 95% Time: 240 hrs
Characteristic (RTC) PPM/°C	-6.0	50	MIL-STD-202-304 Hot Cycle: 25°, 50°, 75°, 125°C Cold Cycle: 25°, 0°, -25°, -55°C
Thermal Shock (Δ R%)	0.2	-0.5	MIL-STD-202-107B No of Cycles: 25 Hot Cycle Temp: 125 °C Cold Cycle Temp: -65 °C
Solder Float (Δ R%) After 1 Cycle After 5 cycles	-0.4 -0.6	0.5	MIL-STD-202-210D Temp: 260°C Immersion: 20 Second
Power Density (mW/mil ²) derated at 50%	0.45	0.15	Step-up Power Test Resistor size 0.020" x 0.030"

Synergistic Effect !

Resistor Sizing



$$L1 = W1$$

$$R1 = 25 \text{ Ohms}$$



$$L2 = W2$$

$$R2 = 25 \text{ Ohms}$$



$$L3 = W3$$

$$R3 = 25 \text{ Ohms}$$



$$3L \times 1W$$

$$R = 75 \text{ Ohms}$$



$$4L \times 2W$$

$$R = 50 \text{ Ohms}$$

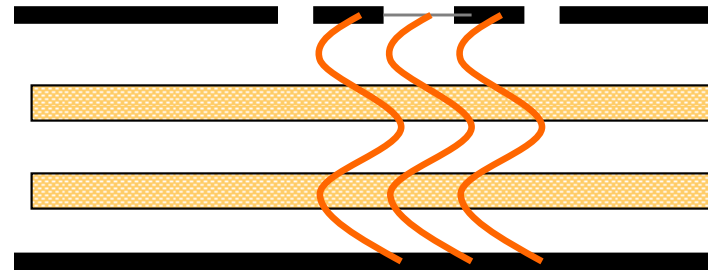
- Shape of Resistor plus Sheet Resistivity determines the Ohmic value
 - Larger L vs. W increases Resistance
- Operating Power determines the size of the Resistor
 - Resistor size goes up as Power requirement goes up
 - Higher Power Density results in smaller Resistors

Performance: Thermal Stability

Buried Capacitance™ Core

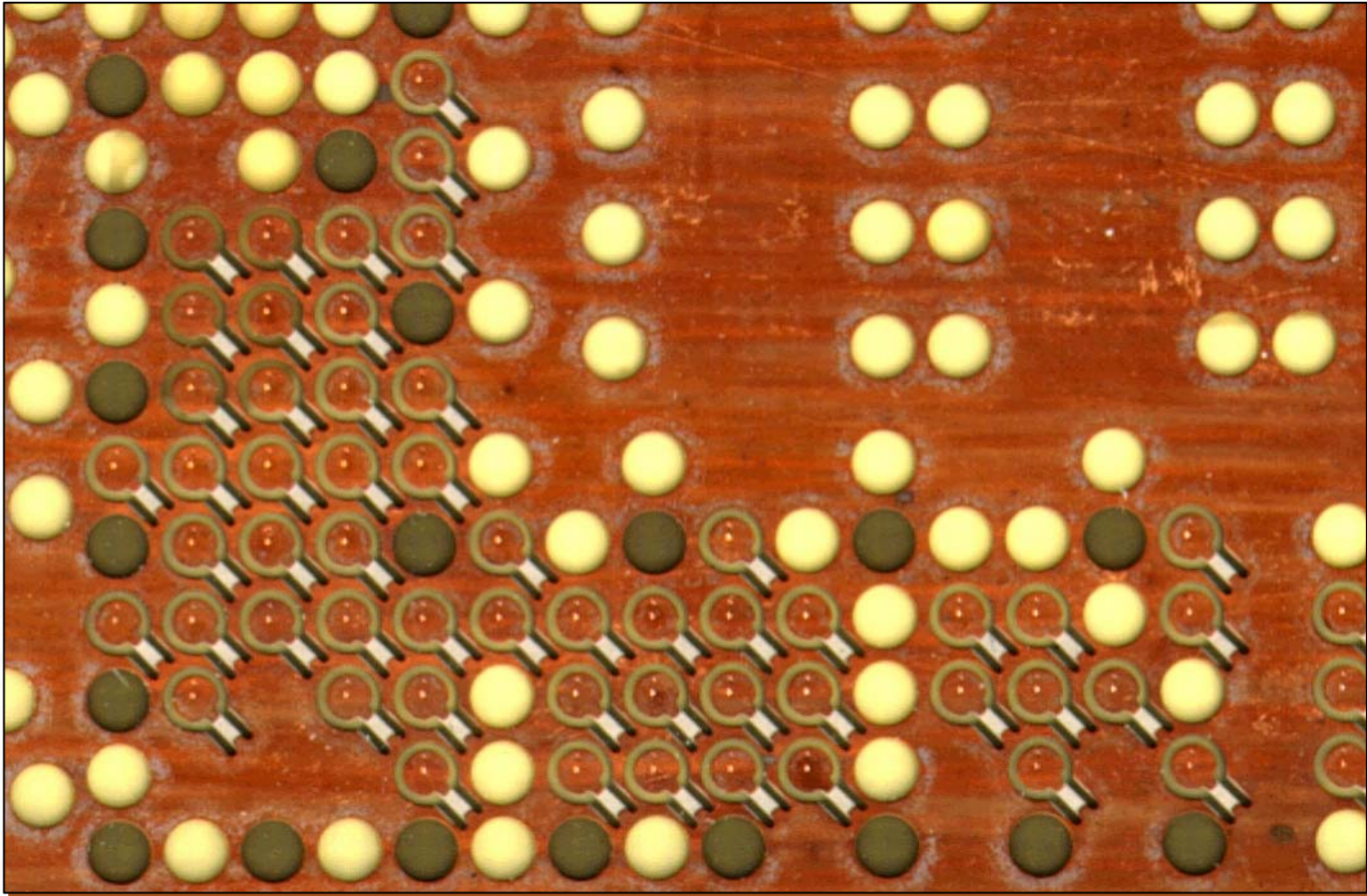


Standard Core



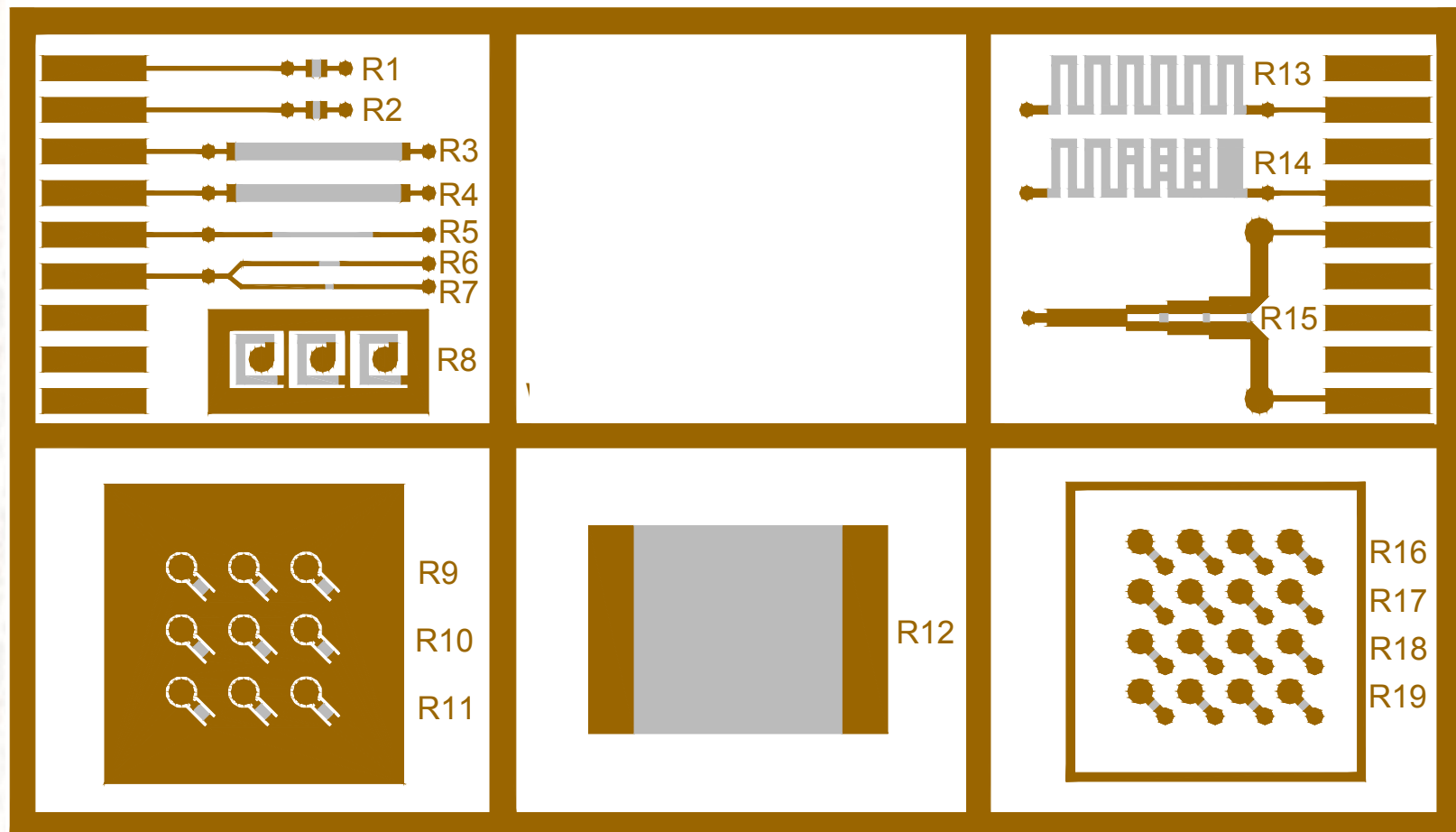
Thinner dielectric provides better heat transfer to copper

Typical Applications



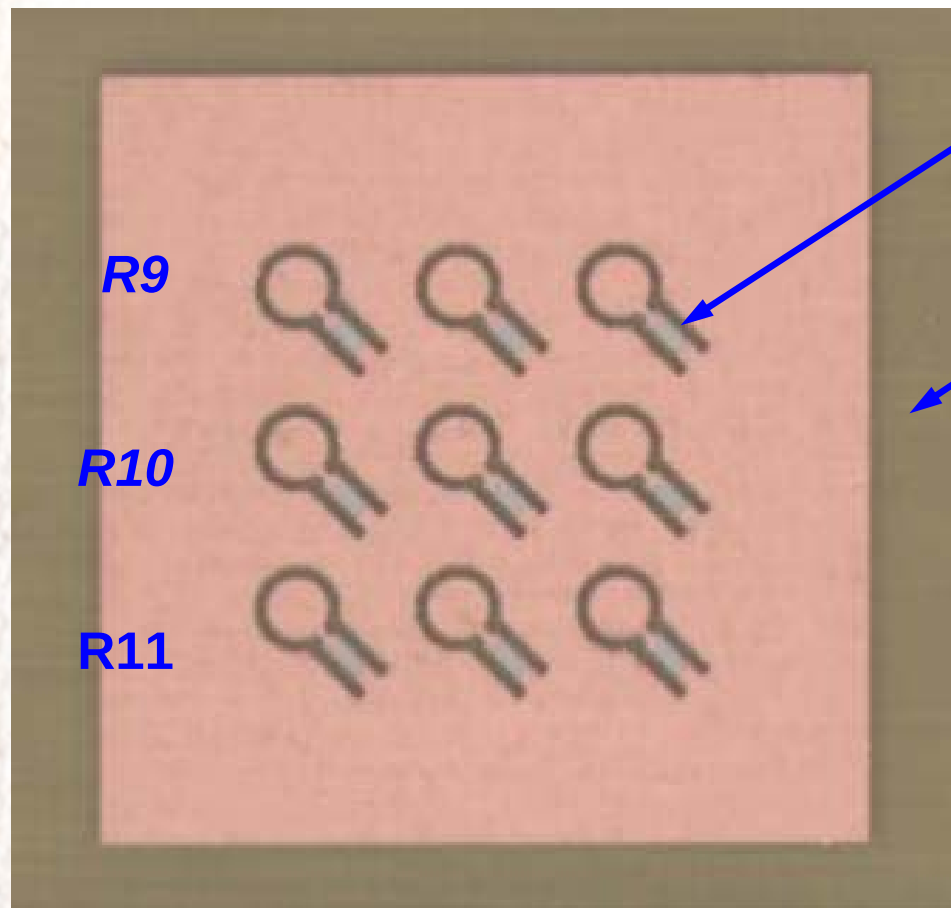
➤ Example of combined product with terminating resistors in an existing layer.

Typical Applications



➤ Example of RC Networks in Demo PCB

Embedded R/C Example



NiP Thin Film Resistors

Capacitor Material

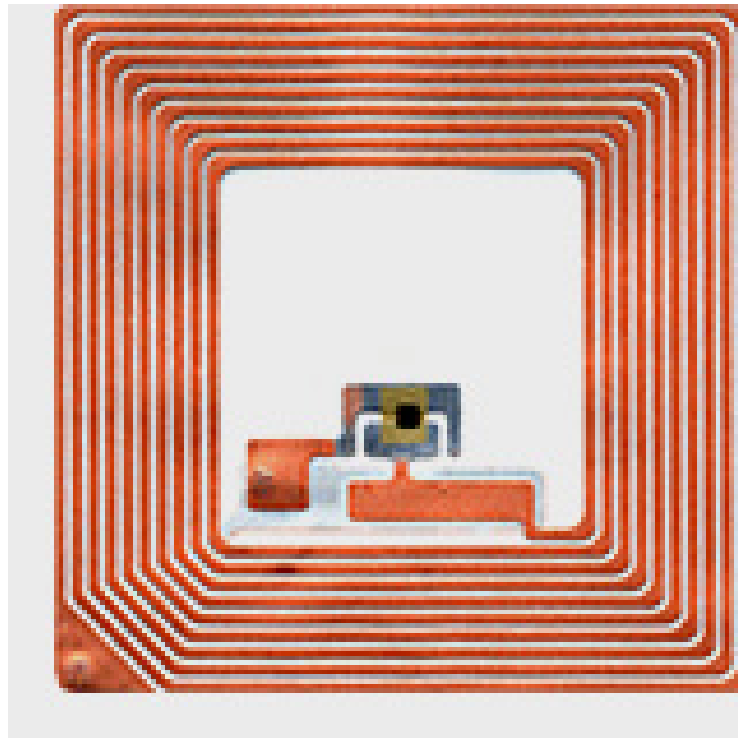
Resistors (R9, R10 & R11)

• **44Ω nominal**

Capacitance (0.6 nF)

• **1.2 nF/in^2**

Possible Application



RFID Tag

Material supplies R/C + L (from Cu on other side)

Summary

Advantages of Combined Embedded R/C Technology

Electrical Advantages

- 1. Improved line impedance matching**
- 2. Shorter signal paths and reduced series inductance**
- 3. Eliminate the inductive reactance of the SMT device**
- 4. Reduced cross talk, noise and EMI**

PCB Design Advantages

- 1. Increase active component density & reduced form factors.**
- 2. Improved wireability due to elimination of via.**
- 3. Improved reliability due to elimination of solder joints.**

Economic Advantages

- 1. Elimination of discrete resistors and capacitors.**
- 2. Improved assembly yield**
- 3. Board densification and/or size reduction**
- 4. Improved Material Cost Structure**

Summary

Advantages of Combined Embedded R/C Technology



Ultimate Lead-Free Solution!

Future Studies

- **Impedance versus Frequency**
 - **Finish Testing of 25 ohm/sq Resistor Foil on 24μ Capacitor Material**
 - **Effect of Ohm/sq on Capacitor Performance**
- **Resistor/Capacitor Properties on other Capacitor Substrates**
 - **Effect of Thickness and Dk**
- **Verify/Quantity Synergistic Effects**
- **Investigate Unique Circuit Patterns**

Conclusion

- **Combined R/C Material results in reliable embedded passives on a single embedded core.**
- **Meets all Resistor Foil Specifications.**
- **Meets all the Buried Capacitance™ Material Specifications.**
- **Any PCB manufacturer capable of making Buried Resistors and Licensed to Produce Buried Capacitance™ cores can produce the Combined R/C PCBs.**
- **Cost advantage of combined product due to reduced form factors, layer count and elimination of greater numbers of SMTs.**
- **Combining the two materials enhances the performance of each material.**
- **Allows unique circuit designs**

Note: Buried Capacitance™ is a trademark of HSCI

**For Further Information visit our
Web Sites**

www.oakmitsui.com

www.ohmega.com